Cervical cancer mortality trends following HPV vaccination in England, 2001–24: an analysis of population-based mortality data

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Summary

Background **Human papillomavirus (HPV) vaccination is a key component of global cervical cancer elimination strategies. While substantial declines in cervical cancer incidence have been observed in several countries, evidence of its effect on mortality is scarce. England introduced a national HPV vaccination programme in 2008 for girls aged 12–13 years (achieving 80–90% coverage before the COVID-19 pandemic), with a catch-up campaign in 2008–10 for girls aged 14–18 years. We aimed to investigate trends in cervical cancer mortality and to estimate the reduction in cervical cancer deaths in young women in England following the introduction of HPV vaccination.**

Methods **We analysed population-based cervical cancer mortality data from England between 2001 and 2024 among women aged 20–24, 25–29, and 30–34 years. HPV vaccination coverage by birth cohort was obtained from official reports and used to estimate, for each age group and calendar year of death, the proportion of women who had been vaccinated. We plotted observed and adjusted age-specific mortality rates against year of death and overlaid a smoothed curve. Poisson regression applied to the numbers of deaths was used to estimate the relative risk reduction in vaccinated women compared with what was expected in the absence of vaccination, assuming no herd immunity. Confidence intervals were obtained by inverting the likelihood ratio test.**

Findings **In women aged 20–24 years between 2020 and 2024, in whom vaccination coverage was around 88–90% at age 12–13 years, no deaths occurred, compared with 23·1 expected deaths based on historical rates, corresponding to a mortality reduction of 100% (95% CI 84–100). In earlier birth cohorts, who were offered vaccination up to age 18 years with coverage of around 63–87%, mortality reductions of 80% (51–94) in women aged 20–24 years in 2015–19, and 69% (55–79) in women aged 25–29 years in 2020–24 were observed. The relative risk reduction in vaccinated women was estimated from population-level data to be 100% (95% CI 81 to 100) in women aged 20–24 years, 100% (89 to 100) in those aged 25–29 years, and 63% (–13 to 100) in those aged 30–34 years. Up until the end of 2024, HPV vaccination in England was associated with a reduction of around 199·6 cervical cancer deaths (95% CI 125·0–274·2).**

Interpretation **Our findings provide the first robust national-level evidence, albeit observational, that high HPV vaccination coverage is associated with a substantial reduction in cervical cancer deaths. This is shown by the substantial decrease in cervical cancer deaths observed among women aged 20–29 years in England, particularly among those vaccinated at ages 12–13 years. These findings support the achievability of the WHO goal of eliminating cervical cancer as a public health problem, and efforts should be made to achieve high vaccine uptake among young adolescents globally.**

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Introduction

Human papillomavirus (HPV) vaccination is one of the three pillars of WHO's global strategy for the elimination of cervical cancer[1] and is currently included in national immunisation programmes in around 150 countries worldwide.[2] Several countries have reported substantial falls in cervical cancer incidence following HPV vaccination,[3–9] but there is little evidence of the effect of HPV vaccination on cervical cancer mortality. One research letter reported a steep decline in cervical cancer mortality in the USA in women aged younger than 25 years: 31 deaths occurred between 2016 and 2021, which was 26 fewer than the number projected from past rates.[10]

A school-based HPV vaccination programme has been in place in England since September, 2008.[11] The routine programme targets girls aged 12–13 years, with boys of the same age included since 2019. A catch-up programme to vaccinate girls aged 14–18 years was implemented in 2008–09 and 2009–10. The high vaccination coverage (of around 90% for one dose before the COVID-19 pandemic) resulted in a large decrease of up to 87% in cervical cancer incidence in cohorts targeted by vaccination, particularly in those targeted

Research in context

Evidence before this study

Human papillomavirus (HPV) vaccination has been shown in randomised controlled trials to prevent persistent HPV infection and high-grade cervical intraepithelial neoplasia. Observational studies have also provided robust evidence that, in real-world settings, cervical cancer incidence has been reduced both in women vaccinated against HPV and in cohorts with high levels of HPV vaccine uptake. However, to date, very little empirical data have shown that HPV vaccination reduces deaths from cervical cancer. We used ChatGPT to search for "papers that report reduced cervical cancer mortality following HPV vaccination". It yielded one research letter on trends in cervical cancer mortality in the USA in women aged 20–24 years, two studies linking HPV vaccination to reduced cervical cancer incidence, and two modelling studies that predict mortality reductions. We then searched PubMed for papers citing the research letter. This search returned ten papers, none of which provided further evidence on cervical cancer mortality following HPV vaccination. An additional search in PubMed using "(HPV vaccination) AND (cervical cancer) AND (mortality OR deaths) AND rates" restricting to "clinical study" or "clinical trial" or "observational study" identified 15 papers; however, these papers did not report on the observed impact of HPV vaccination on cervical cancer mortality.

Added value of this study

This study shows substantial reductions in cervical cancer mortality in England among women aged 20–29 years following the introduction of the national HPV vaccination programme. The lower limit of the 95% CI for the relative reduction in the risk of death from cervical cancer in vaccinated women is over 80% in the 20–24 years and 25–29 years age groups. No deaths from cervical cancer among women aged 20–24 years occurred in England during the last 5 years of our study follow-up (2020–24).

Implications of all the available evidence

These data suggest that HPV vaccination programmes for adolescent females can lead to reduced deaths from cervical cancer, reinforcing the importance of achieving and maintaining high vaccination coverage, particularly in females aged 12–15 years (while most are still sexually inactive).

before age 16 years.[5,9,12] However, until now, the effect of vaccination on cervical cancer mortality has not been empirically assessed.

We aimed to analyse trends in cervical cancer mortality in England to investigate the effect of the HPV vaccination programme on deaths from cervical cancer.

Methods

Mortality and population data

See Online for appendix

Mortality data for England for the years 2001–23 were downloaded from the National Disease Registration Service open dashboard,[13] and data for 2024 from Nomis (a service provided by the Office for National Statistics [ONS]).[14] The dashboard does not allow data to be subdivided by race or ethnicity. Additional ONS mortality data for 2000 were retrieved from the National Archives,[15] but were available only for England and Wales combined. Mid-year population estimates for 2000–24 required for the analysis were also obtained from Nomis.[16] Cervical cancer deaths were identified using ICD codes for malignant neoplasm of the cervix uteri: in the 2000 data, we used ICD-9 code 180, while those registered from 2001 onwards were identified using ICD-10 code C53. ONS population estimates for females are based on sex (registered at birth) as self-reported in the census.

Neither ethics committee (institutional review board) approval nor informed consent were required because this research uses only data that are already public and de-identified, and thus the study does not constitute research on human participants.

HPV vaccination data

The first birth cohort eligible for HPV vaccination within the national programme comprises females born between September, 1990 and August, 1991. Those individuals were aged 20 years in 2010–11 and were aged 33–34 years in 2024. Academic year-specific HPV vaccination coverage data were extracted from the UK Health Security Agency's annual reports.[17] Coverage data were available by birth cohort defined using academic years, with each cohort spanning from Sept 1 to Aug 31 of the following year. These data were combined to derive calendar year-specific coverage. Details are provided in the appendix (p 1). National data were not available by race or ethnicity.

For the purposes of HPV vaccination, sex was determined using National Health Service records and school registers.

Statistical analysis

For each calendar year and age group, mortality rates were calculated as the number of deaths divided by the corresponding women-years, with the latter approximated using mid-year population estimates. For tabulation purposes, we grouped these counts and rates in 5-year intervals.

We also calculated adjusted mortality rates by dividing the number of deaths by the women-years corresponding to unvaccinated women. These rates correspond to those in unvaccinated women in the hypothetical scenario in which all the cervical cancer deaths are confined to unvaccinated women (because vaccinated women are assumed to have zero risk).

We plotted the observed age-specific mortality rates against year of death and superimposed a smooth fit to these data and a bar chart of the age-specific and year-specific proportion of women-years vaccinated against HPV. For the crude and adjusted rates, the smooth fit was estimated with a restricted cubic spline[18] with four degrees of freedom in year of death, using Poisson regression with robust standard errors.[19]

We modelled the number of deaths observed in each year and each 5-year age group using a Poisson distribution, with the expected count equal to the rate multiplied by the relevant women-years at risk. The rate was modelled separately in each age group as $r_U(1-p)+\theta r_U p$, where r_U is the mortality rate in unvaccinated women, θ is the relative risk in vaccinated women, and p is the proportion of the population who received at least one dose of HPV vaccine (ie, coverage). Hence, $1-\theta$ represents the relative risk reduction (RRR). We fit the model by maximum likelihood. We also fit two special versions corresponding to θ=0 (perfect RRR) and θ=1 (no RRR). When θ=0, the rate is equal to $r_U(1-p)$ and r_U can be estimated using standard Poisson regression with the logarithm of the women-years of unvaccinated women as an offset. When θ=1, the rate simplifies to r_U, which can be estimated using standard Poisson regression with the usual log(women-years) (ie, vaccinated and unvaccinated women combined) as an offset. In each age group, we modelled the logarithm of r_U to be linear in the year of death—ie, $\log(r_U)=a+bT$, where a and b are parameters (allowed to be different across age groups) and T refers to the year of death. To ensure the estimate of θ was not negative, we estimated $g=\log(\theta)$ and then back-transformed to the original scale using $\theta=\exp(g)$. 95% CIs for θ were estimated by inverting the likelihood ratio statistic, assuming that it is distributed χ^2 on one degree of freedom, corresponding to a threshold of 3·841. We used the threshold of 3·841 for the upper value even if the lower value was θ=0.

We compared model fits using the log-likelihood and the Pearson goodness-of-fit statistic.

We also estimated the number of cervical cancer deaths prevented up to the end of 2024 among cohorts offered vaccination. First, we derived the age-specific and year-specific counterfactual rates (r_U), defined as the rates expected in the absence of vaccination. These rates were then multiplied by the corresponding population in each year and summed over all the relevant years in that age group. The number of deaths prevented is the difference between the counterfactual sum and the observed total number of cervical cancer deaths. We approximated the variance of the difference by treating the (robust) variances of the counterfactuals in each age group and the (Poisson) variance of the observed number of deaths as independent.

All analyses were done in Stata version 18. Models were fitted using the poisson and the mlexp commands. The counterfactual number of deaths and its variance were estimated using nlcom.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

The number of deaths and the rates of cervical cancer mortality in the three age groups (20–24, 25–29, and 30–34 years) are shown for 5-year calendar periods between 2000 and 2024 in figure 1, with shading indicating level of HPV vaccination coverage. The highest level of vaccination coverage shown corresponds to around 90% of the population receiving at least one dose of HPV vaccine at age 12–13 years or as part of catch-up programmes in school years 9 and 10, while the lowest (excluding 0% coverage) corresponds to between 1% and 50% coverage, with most of the vaccination having

	Age 20–24 years			Age 25–29 years			Age 30–34 years		
	Deaths	Women-years (millions)	Rate per 100 000 women-years	Deaths	Women-years (millions)	Rate per 100 000 women-years	Deaths	Women-years (millions)	Rate per 100 000 women-years
Year of death									
2000*–04	25	7·72	0·32	59	8·22	0·72	171	9·59	1·78
2005–09	16	8·49	0·19	88	8·63	1·02	141	8·57	1·65
2010–14	27	8·95	0·30	123	9·22	1·33	145	9·06	1·60
2015–19	5	8·95	0·06	73	9·63	0·76	140	9·75	1·44
2020–24	0	8·54	0·00	32	9·59	0·33	131	10·33	1·27

Vaccination coverage
- None
- 1–50%
- 51–87%
- 88–91%

Figure 1: **Age-specific mortality counts, women-years, and crude rates for 2000–24 by 5-year period of death**
Shading indicates HPV vaccination coverage (appendix p 1). HPV=human papillomavirus. *Data for 2000 refer to England and Wales combined and are shown here for completeness only, but were not used in the regression analysis; note that since the population of Wales is about 5·5% of that of England, the death counts from Wales in 2000 are expected to be approximately one death in women aged 20–29 years (combined) and two in women aged 30–34 years.

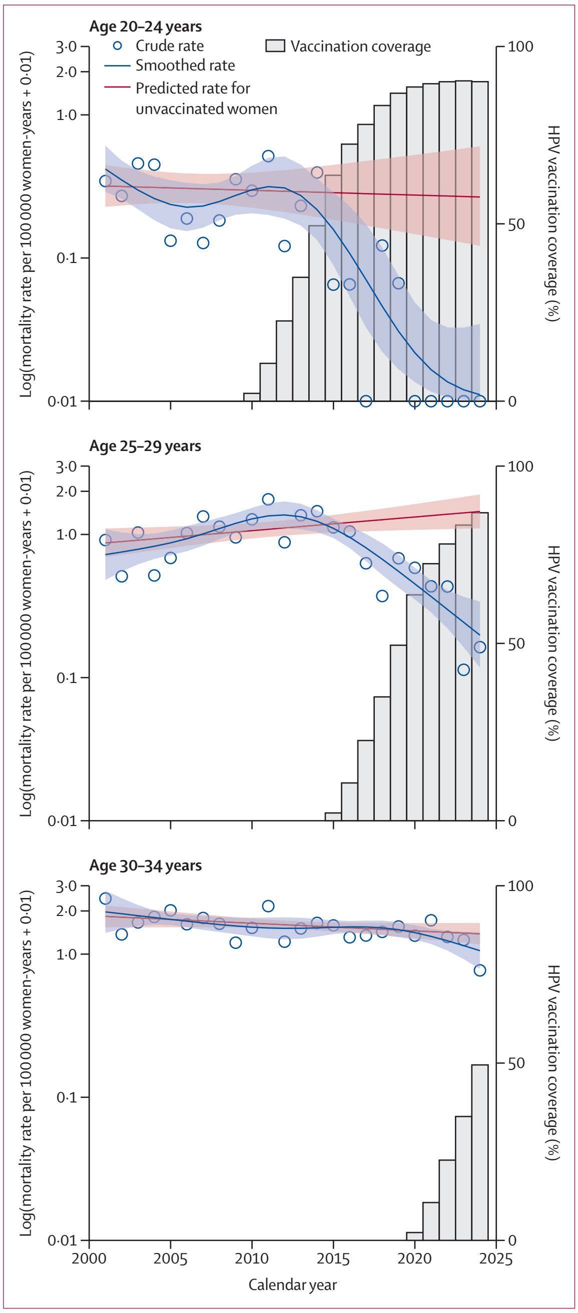

Figure 2: **Age-specific scatter plots of crude mortality rates by year of death, including smoothed curves and predicted rates for unvaccinated women**
Rates are displayed on a log scale after adding a small constant (since the logarithm of zero is undefined). Shaded regions are 95% CIs for the smoothed and predicted mortality rates. Bars show the proportion of women (in that age group and in that calendar year) who had been vaccinated against HPV during adolescence. In each age group, vaccination in the first 4 years of non-zero coverage was done when the recipients were aged 15–18 years and might therefore already have been exposed to HPV, and thus the effectiveness of vaccination is expected to be lower in these years. The smooth curve was obtained using a simple Poisson regression model with a restricted cubic spline for calendar year and no adjustment for vaccine coverage. The predicted rates in unvaccinated women were derived from the Poisson regression model used to estimate the relative risk reduction in vaccinated women. HPV=human papillomavirus.

occurred at age 14–18 years (see appendix p 1 for details of vaccine coverage). Notably, there were no deaths from cervical cancer in women aged 20–24 years in 2020–24, compared with 23·1 expected deaths based on the average rate (0·27 deaths per 100 000 women-years) observed between 2000 and 2014—a reduction in cervical cancer mortality of 100% (95% CI 84–100). Compared with the 2000–14 period, mortality rates decreased by 80% (95% CI 51–94) among women aged 20–24 years in 2015–19, and by 69% (55–79) among women aged 25–29 years in 2020–24. Since a proportion of women aged 20–24 years in 2010–14 will have been vaccinated—a mean of 24%—these descriptive reductions are conservative.

Results by individual year of death are plotted in figure 2. Notably, the rates start reasonably constant (they decline slightly in women aged 20–24 years and 30–34 years, and increase slightly in women aged 25–29 years) and decline sharply once vaccine coverage becomes substantial. It should be noted that in women aged 30–34 years, not only were a smaller proportion vaccinated (under 50% even in 2024), but most of those will have been vaccinated at age 15–18 years when they might have already been infected by HPV16 and/or HPV18, and thus the vaccine will have been less effective.

The table presents the results of fitting the Poisson regression models in each age group. In the 20–24 years and 25–29 years age groups, the RRRs in those vaccinated are estimated to be 100% (95% CI 81–100) and 100% (89–100), respectively, corresponding to very large negative values for the estimated logarithm of the relative risk. In women aged 30–34 years, the RRR is estimated to be 63% (–13 to 100; ie, ranging from a relative increase in mortality in vaccinated women up to a maximum relative reduction of 100%). The reason the estimated RRR could not be greater than 100% in vaccinated women is because the model forces no reduction in unvaccinated women. Without this restriction, in the presence of herd protection in unvaccinated women, the model might estimate the risk in vaccinated women to be negative so as to better approximate the observed risk of the population of vaccinated and unvaccinated women combined. For women aged 20–24 years and 25–29 years, the p values for testing whether the vaccine is effective (rejecting the null hypothesis of θ=1) are less than 0·001.

We estimated that in England, up to the end of 2024, the HPV vaccination programme was associated with a reduction of around 199·6 cervical cancer deaths (95% CI 125·0–274·2).

Figure 2 shows that the model predicts increasing rates of cervical cancer mortality in the absence of vaccination in women aged 25–29 years. As a sensitivity analysis in women aged 25–29 years, we also fitted the model in which the mortality rate in unvaccinated women is constrained to be constant over time. In that model, the estimated RRR in vaccinated women is 94% (95% CI 78–100).

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	Age 20–24 years			Age 25–29 years			Age 30–34 years		
	No RRR	Estimated RRR	Perfect RRR	No RRR	Estimated RRR	Perfect RRR	No RRR	Estimated RRR	Perfect RRR
RRR* (95% CI)	0%	100% (81 to 100)	100%	0%	100% (89 to 100)	100%	0%	63% (–13 to 100)	100%
Pearson goodness-of-fit	40·59	26·67	26·67	77·85	31·33	31·33	21·66	19·41	21·22
Log likelihood	–46·79	–41·16	–41·16	–93·00	–69·23	–69·23	–73·22	–71·82	–72·64
p value†	··	0·0008	··	··	<0·0001	··	··	0·094	··

The no RRR columns show values when the relative risk (θ) is set at 1, corresponding to the null hypothesis that there is no effect of HPV vaccination on cervical cancer mortality. The perfect RRR columns show values when θ is set at 0, corresponding to the alternative hypothesis that women vaccinated against HPV have no risk of dying from cervical cancer (ie, the vaccination works perfectly). RRR=relative risk reduction. *RRR in cervical cancer mortality in vaccinated women, defined as 1 – θ, expressed as a percentage; θ in vaccinated women is estimated assuming no herd protection in unvaccinated women. †H0: no effect of HPV vaccination (θ=1).

***Table:* Estimated RRR in cervical cancer in vaccinated women and model fit in each age group**

When plotting the adjusted rates (dividing the number of deaths by women-years in unvaccinated women), the rates change little over time in women aged 25–29 years, suggesting that the model with 100% RRR in vaccinated women fits well (appendix p 2). In women aged 20–24 years, there is still a downwards trend due to there being zero deaths in recent years, but the 95% CIs show that the data are consistent with the mortality rates in unvaccinated women being constant over time. The wide confidence intervals are a function of the small number of unvaccinated women since 2020. In women aged 30–34 years, the adjusted rate shows a slight increase since 2020, suggesting that the relative reduction in cervical cancer mortality associated with HPV vaccination (primarily at age 15–18 years) is less than 100%.

Discussion

Remarkably, this study reports no deaths from cervical cancer in women aged 20–24 years in England in the five most recent years of available data (2020–24). That is likely to be a chance finding corresponding to a very low underlying rate of cervical cancer mortality in the population, rather than a true eradication of cervical cancer as a cause of death in young women. Nevertheless, the substantial reductions in cervical cancer mortality observed in women aged 20–24 years and 25–29 years in recent years are consistent with perfect vaccination effectiveness against cervical cancer mortality in vaccinated women (particularly so in those vaccinated aged 12–13 years). HPV vaccination effectiveness against cervical cancer mortality is unlikely to be 100%, but it is likely that the small risk in vaccinated women is compensated by a reduction of risk in unvaccinated women thanks to herd protection.[20] Since the HPV vaccine offered to those born before 2000 was bivalent, it is unlikely to have provided more than 90% protection against cervical cancer mortality below age 30 years in vaccinated women; a study found that in a series of 98 cervical cancers in women younger than 30 years in England, 91 (93%) had HPV16 or HPV18.[21]

Whereas we and others have previously shown a reduction in cervical cancer incidence associated with vaccination,[3–9] there might still have been a lingering concern that the cancers prevented by vaccination were the same as those that would previously have been detected at stage I through screening, and that the advanced-stage cancers in unscreened women that account for most of the deaths would not be affected by population vaccination. In the current analysis, we have shown that a substantial decrease in cervical cancer mortality has followed the national HPV vaccination programme and that the decline in mortality is as, or better than, predicted.

It is too early to observe conclusive evidence of the effect of HPV vaccination on cervical cancer mortality in women aged 30–34 years, but the data are consistent with a substantial effectiveness even when delivered to older teenagers (aged 14–18 years).

The estimated nearly 200 cervical cancer deaths prevented by HPV vaccination up to the end of 2024 are likely to represent only a small fraction of the potential benefit; as the cohort of vaccinated women ages, the number of deaths prevented is expected to increase exponentially, at least for the next two decades.

Few previous studies have explicitly analysed the association between HPV vaccination status or HPV vaccination uptake and cervical cancer mortality. A research letter by Dorali and colleagues[10] with data up to 2021 reported that cervical cancer mortality in the USA among women younger than 25 years, which had been falling steadily between 1992–94 and 2013–15, showed an accelerated decline up to 2019–21. However, the authors did not provide detailed data on HPV vaccination coverage in the USA, and did not attempt to estimate the effect of HPV vaccination on cervical cancer mortality in vaccinated young women. It was reported that in 2014, coverage of at least one HPV vaccine dose in the USA among females aged 13–17 years was 60%,[22] so the fall in mortality reported by Dorali and colleagues[10] is consistent with mortality reducing by the same amount as the coverage. In a recent paper discussing trends in the incidence of grade 3 cervical intraepithelial neoplasia (CIN3), the incidence of cervical cancer, and the mortality from cervical cancer in England using national data up to 2022, the authors commented on the substantial reduction in CIN3 and cervical cancer in women aged

20–24 years and 25–29 years associated with HPV vaccination, but did not comment on any association between vaccine coverage and mortality.[23]

The major strength of our study is that it uses national data on mortality and HPV vaccination coverage. Additionally, the use of an extended Poisson regression model that is not linear in its parameters allows for the estimation of the protective effect of vaccination in those vaccinated, despite only having ecological data. The limitations are that the analysis relies on population-level coverage statistics rather than individual-level vaccination status, and that we only had access to mortality in 5-year age groups. Additionally, the model assumes no herd protection among those not vaccinated. Had we been able to access data corresponding to single school years (September to August) of birth, the association between cervical cancer mortality and HPV vaccine uptake would have been more precise. With the available data, we were only able to estimate the RRR in vaccinated women by making strong assumptions about the independence of vaccination uptake and underlying risk, and a lack of herd protection in those not vaccinated within the birth cohort.

Whereas it would normally be unwise to make causal inference from ecological data, we believe that combining external evidence and using the Bradford Hill criteria,[24] we can infer that HPV vaccination has led to a marked reduction in cervical cancer mortality in England. That evidence includes randomised controlled trials showing that vaccination in young women prevents persistent HPV infection and high-grade cervical intraepithelial neoplasia.[25,26] Observational data also show reduced rates of cervical cancer in vaccinated compared with unvaccinated women within the same birth cohort,[3,4,27] and a dose–response relationship between the HPV vaccination uptake and the reduction in cervical cancer incidence in birth cohorts.[5,8,9] Furthermore, confounding from changes to cervical screening,[12] treatment, or sexual behaviour is unlikely to account for more than a small proportion of the observed changes in cervical cancer mortality. The primary changes to cervical screening in England that might affect these birth cohorts are cessation of screening under the age of 24·5 years and falling screening coverage—both of which would, if anything, lead to an increase in cervical cancer mortality—and a switch from cytology to HPV testing from 2019, which is unlikely to have had a substantial effect on mortality in 2020–24. Changes in sexual behaviour are also unlikely to account for the observed falls in cervical cancer mortality because there were only modest fluctuations in the rates of new sexually transmitted infections in England between 2015 and 2023.[28] Finally, any change in treatment of frank invasive cervical cancers would be expected to affect all ages at the same time, whereas the decreases in mortality in our data occurred 5 years later in each successive 5-year age group.

The strong dose–response relationship in our observational data further supports the causal argument. It is to be expected that vaccination of older teenagers will be less effective because they are more likely to have been infected with HPV before vaccination. Furthermore, for each calendar year of death, we estimated the proportions of women aged 20–24 years and 25–29 years who had been vaccinated (as adolescents), showing that the magnitude of the reduction in rates was highly correlated with those proportions and that the year in which rates are seen to begin to fall sharply coincides with the year in which, for the first time, a substantial proportion of the at-risk population were vaccinated as adolescents.

In this study, using national data on HPV vaccination uptake between 2008–09 and 2017–18 and mortality data for England from 2001 to 2024, we found a substantial reduction in cervical cancer mortality in women aged 20–29 years that was associated with high uptake of HPV vaccination. This observation further supports the benefit of HPV vaccination in reducing not only the incidence of but also the mortality from a cancer that is globally still the second most common cause of cancer death in women younger than 65 years.[29] With falling HPV vaccine rates and growing vaccine hesitancy, it is important to demonstrate, as we have done here, that following the introduction of HPV vaccination with high population coverage, there is a considerable fall in cervical cancer mortality.

Contributors

PS conceptualised the study and drafted the manuscript. PS and MF developed the statistical methods and wrote the Stata code for the data analysis. MF downloaded and curated the data and reviewed the manuscript. Both authors directly accessed and verified the underlying data reported in the manuscript and approved the final submitted version. PS was responsible for the decision to submit the manuscript for publication.

Declaration of interests

PS is a lead investigator on a trial studying the HPV vaccine Gardasil-9 that was supported in part by a research grant from the Investigator-Initiated Studies Program of Merck Sharp & Dohme. He was a member of the data safety monitoring board for the IARC India HPV Vaccine Trial. PS and MF received funding from RM Partners (NW & SW London Cancer Alliance) for the evaluation of an intervention aimed at increasing HPV vaccination uptake.

Data sharing

All data are publicly available, as reported in the Methods.

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References

1 WHO. Global strategy to accelerate the elimination of cervical cancer as a public health problem. 2020. https://www.who.int/publications/i/item/9789240014107 (accessed Jan 29, 2026).

2 Han J, Zhang L, Chen Y, et al. Global HPV vaccination programs and coverage rates: a systematic review. *EClinicalMedicine* 2025; **84:** 103290.

3 Lei J, Ploner A, Elfström KM, et al. HPV vaccination and the risk of invasive cervical cancer. *N Engl J Med* 2020; **383:** 1340–48.

4 Kjaer SK, Dehlendorff C, Belmonte F, Baandrup L. Real-world effectiveness of human papillomavirus vaccination against cervical cancer. *J Natl Cancer Inst* 2021; **113:** 1329–35.

5 Falcaro M, Castañon A, Ndlela B, et al. The effects of the national HPV vaccination programme in England, UK, on cervical cancer and grade 3 cervical intraepithelial neoplasia incidence: a register-based observational study. *Lancet* 2021; **398:** 2084–92.

6 Palmer TJ, Kavanagh K, Cuschieri K, et al. Invasive cervical cancer incidence following bivalent human papillomavirus vaccination: a population-based observational study of age at immunization, dose, and deprivation. *J Natl Cancer Inst* 2024; **116:** 857–65.

7 Middeldorp M, Brouwer JGM, Duijster JW, et al. The effect of bivalent HPV vaccination against invasive cervical cancer and cervical intraepithelial neoplasia grade 3 (CIN3+) in the Netherlands: a population-based linkage study. *Lancet Reg Health Eur* 2025; **54:** 101327.

8 Wu S, Deng Y, Lepp T, et al. Extended follow-up of invasive cervical cancer risk after quadrivalent HPV vaccination: nationwide, register based study. *BMJ* 2026; **392:** e087326.

9 Falcaro M, Soldan K, Ndlela B, Sasieni P. Effect of the HPV vaccination programme on incidence of cervical cancer and grade 3 cervical intraepithelial neoplasia by socioeconomic deprivation in England: population based observational study. *BMJ* 2024; **385:** e077341.

10 Dorali P, Damgacioglu H, Clarke MA, et al. Cervical cancer mortality among US women younger than 25 years, 1992–2021. *JAMA* 2025; **333:** 165–66.

11 UK Health Security Agency. Human papillomavirus (HPV): the green book, chapter 18a. 2023. https://www.gov.uk/government/publications/human-papillomavirus-hpv-the-green-book-chapter-18a (accessed Feb 5, 2024).

12 Falcaro M, Castañón A, Ndlela B, Sasieni P. Association between HPV vaccination and cervical screening policy changes and cervical cancer incidence and grade-3 cervical intraepithelial neoplasia incidence in England, 2006–2020: a population-based trends analysis. *Lancet Reg Health Eur* 2024; **49:** 101157.

13 National Disease Registration Service. Cancer incidence and mortality. 2025. https://nhsd-ndrs.shinyapps.io/incidence_and_mortality/ (accessed Jan 26, 2026).

14 Nomis, Office for National Statistics. Mortality statistics—underlying cause, sex and age. Oct 9, 2025. https://www.nomisweb.co.uk/datasets/mortsa (accessed Jan 26, 2026).

15 Office for National Statistics. The 20th century mortality files, 1901–2000. March 11, 2011. https://webarchive.nationalarchives.gov.uk/ukgwa/20150908090558/http://www.ons.gov.uk/ons/publications/re-reference-tables.html?edition=tcm%3A77-215593 (accessed Jan 29, 2026).

16 Nomis, Office for National Statistics. Population estimates—local authority based by single year of age. 2025. https://www.nomisweb.co.uk/datasets/pestsyoala (accessed Jan 26, 2026).

17 UK Health Security Agency. HPV vaccine uptake. 2026. https://www.gov.uk/government/collections/vaccine-uptake#hpv-vaccine-uptake (accessed March 3, 2026).

18 Durrleman S, Simon R. Flexible regression models with cubic splines. *Stat Med* 1989; **8:** 551–61.

19 Cameron AC, Trivedi PK. Regression analysis of count data, 2nd edn. Cambridge: Cambridge University Press, 2013.

20 Drolet M, Bénard É, Pérez N, et al, and the HPV Vaccination Impact Study Group. Population-level impact and herd effects following the introduction of human papillomavirus vaccination programmes: updated systematic review and meta-analysis. *Lancet* 2019; **394:** 497–509.

21 Mesher D, Cuschieri K, Hibbitts S, et al. Type-specific HPV prevalence in invasive cervical cancer in the UK prior to national HPV immunisation programme: baseline for monitoring the effects of immunisation. *J Clin Pathol* 2015; **68:** 135–40.

22 Reagan-Steiner S, Yankey D, Jeyarajah J, et al. National, regional, state, and selected local area vaccination coverage among adolescents aged 13–17 years—United States, 2014. *MMWR Morb Mortal Wkly Rep* 2015; **64:** 784–92.

23 Gilham C, Peto J. Is elimination of cervical cancer in sight in England? *Prev Med* 2025; **191:** 108218.

24 Hill AB. The environment and disease: association or causation? *Proc R Soc Med* 1965; **58:** 295–300.

25 Rambout L, Hopkins L, Hutton B, Fergusson D. Prophylactic vaccination against human papillomavirus infection and disease in women: a systematic review of randomized controlled trials. *CMAJ* 2007; **177:** 469–79.

26 FUTURE II Study Group. Quadrivalent vaccine against human papillomavirus to prevent high-grade cervical lesions. *N Engl J Med* 2007; **356:** 1915–27.

27 Wu S, Ploner A, Astorga Alsina AM, Deng Y, Ask Schollin L, Lei J. Effectiveness of quadrivalent human papillomavirus vaccination against high-grade cervical lesions by age and doses: a population-based cohort study. *Lancet Reg Health Eur* 2025; **49:** 101178.

28 UK Health Security Agency. Sexually transmitted infections and screening for chlamydia in England: 2024 report. Dec 4, 2025. https://www.gov.uk/government/statistics/sexually-transmitted-infections-stis-annual-data-tables/sexually-transmitted-infections-and-screening-for-chlamydia-in-england-2024-report#overall-trends (accessed April 14, 2026).

29 International Agency for Research on Cancer. Global Cancer Observatory. 2026. https://gco.iarc.fr/ (accessed April 14, 2026).